

Detection and Manipulation of Semi-Occluded Objects

Barrel detection in 3D point clouds — benchmarking LiDAR-only and camera-LiDAR fusion methods

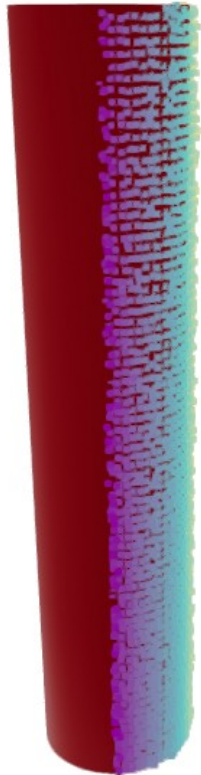
Master's thesis · Bharath · June 2026

Approach: a method-agnostic benchmark, scaled to simulation

- **Standardised I/O contract:** every detector consumes the same point cloud and emits a parametric cylinder (radius, axis, axis-point, extent) — evaluation is therefore method-agnostic and identical across methods.
- **Stage 1 — establish and validate the metric fit on controlled input:**
 - synthetic clouds with analytic ground truth, plus real depth captures for sim-to-real transfer.
- **Stage 2 — scale ground-truthed data generation with NVIDIA Isaac Sim:**
 - per-barrel pose/size/count, instance & semantic segmentation, RTX LiDAR matched to the real sensor with a co-registered camera; scripted occlusion, burial depth and inter-barrel spacing.
- **Stage 3 — extend the benchmark from LiDAR-only geometric to learned detectors and camera-LiDAR fusion, scored on the same metrics.**
 - Metrics: precision / recall / F1, radius RMSE, axis-angle error, recall vs. occlusion fraction, runtime; annotation/training cost tracked so geometric and learned methods compare fairly.

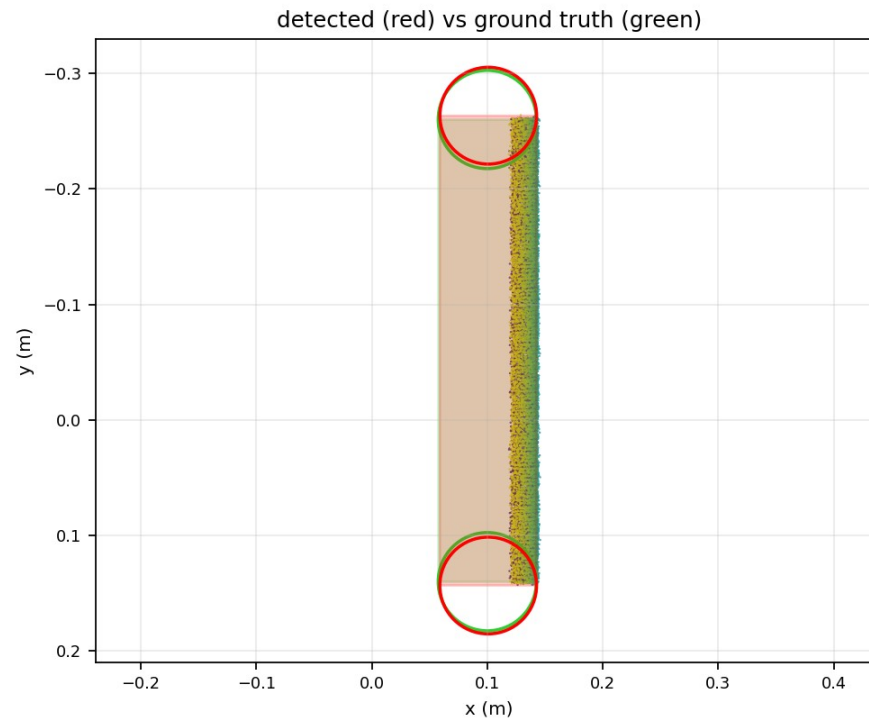
Validation on synthetic clouds with analytic ground truth

Generated cylinders with analytic radial normals — exact (radius, axis, centre) known



3D: fitted barrel + synthetic points

SYNTHETIC DATA — generated cloud with exact ground truth



2D overlay: detected (red) vs ground truth (green)

- Red = detected cylinder, green = ground truth — near-exact overlap.
- Set spans a one-sided $\sim 120^\circ$ arc (Xtion-like partial view) and an additive Gaussian-noise sweep ($\sigma = n0.1-0.3$).
- Top methods: $P = R = F1 = 1.00$; radius RMSE < 0.5 cm; axis-angle error $< 0.5^\circ$.
- Upper bound on fit accuracy where truth is exact, before real-sensor noise.

SYNTHETIC DATA

Transfer to real depth captures

Asus Xtion Pro, single barrel, $\sim 180^\circ$ of the surface visible (one-sided view)



raw depth cloud



fitted barrel (red) + measured points

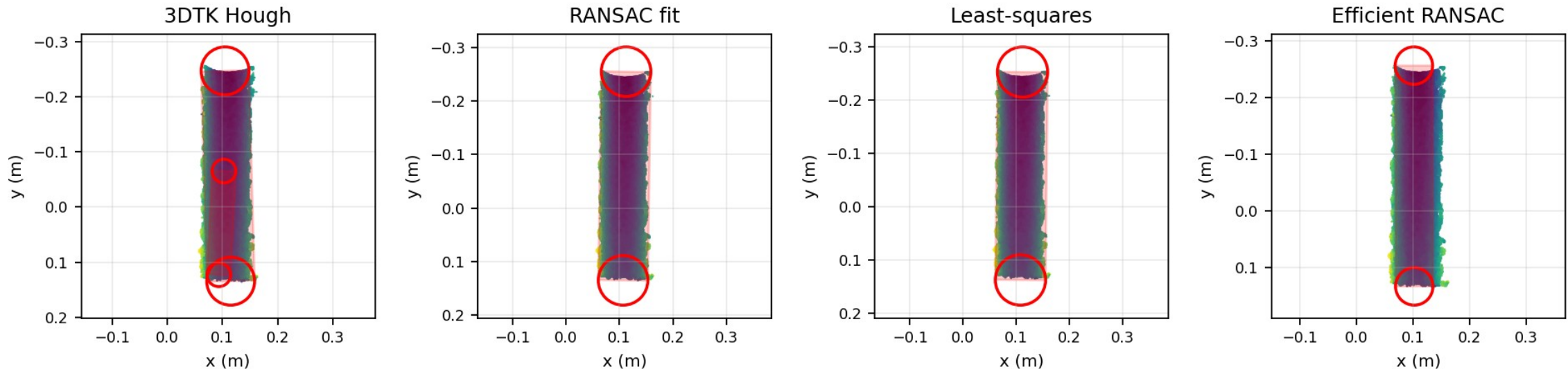
- Pipeline: depth $\rightarrow$ surface-normal estimation (Open3D) $\rightarrow$ cylinder fit.
- RANSAC fit: radius 4.63 cm vs. 4.25 cm physical (error 0.38 cm).
- Axis-angle error 1.85° against the vertical reference.
- Real-sensor radius error $\sim 6-8\times$ the synthetic figure — quantifies the sim-to-real gap.

REAL DATA

Four LiDAR-only geometric detectors on identical input

Same real barrel cloud — four independent algorithms

FOUR DETECTION METHODS on the same REAL barrel capture



- **3DTK randomized 2-step Hough (baseline) • RANSAC primitive fit • nonlinear least-squares cylinder fit • Schnabel Efficient RANSAC (CGAL).**
- Architecture: a clustering proposer feeds a reusable metric fit; Efficient RANSAC is self-contained (no proposer) — isolating the proposer as a confound.
- The baseline emits a phantom secondary cylinder (panel 1 centre) and is non-deterministic; the seeded fitting methods stay clean.

Quantitative accuracy on the real capture

Identical metric applied to every method (axis-angle, radius error vs. known drum)

Method	Radius error	Axis-angle error	Runtime
RANSAC primitive fit	0.38 cm	1.85°	~0.7 s
Nonlinear least-squares	0.47 cm	1.64°	~9 s
Efficient RANSAC (Schnabel)	0.94 cm	0.10°	~0.05 s
3DTK Hough (baseline)	0.09 cm	1.53°	—

- All methods recover radius to ~1 cm and axis to ~2°; operating points differ — Efficient RANSAC fastest, fitting methods most robust, the baseline over-reports (a phantom second cylinder).
- *Single-barrel, low-clutter scene at ~180° visibility — the discriminating regime (occlusion, burial, clutter, multiple instances) is the next experiment.*

Status: implemented vs. pending

Implemented

- Method-agnostic capture → detect → evaluate pipeline with a shared schema
- 4 LiDAR-only geometric detectors (Hough, RANSAC, LS, Efficient RANSAC)
- Validated on synthetic ground truth + first real captures
- Literature survey of ~18 methods across the three families

Pending

- Occlusion sweep: recall and pose error vs. occlusion fraction (the headline experiment)
- Learned detectors (PointNet++ / BarrelNet-style, occlusion-aware BtcDet) — require labelled training data
- Multi-instance and cluttered real scenes (exercises the proposer)
- Camera-LiDAR fusion pipeline (camera-first and LiDAR-first)
- **Validation on a real industrial test-site dataset**

Simulation pipeline & fusion strategy

NVIDIA Isaac Sim — ground-truthed data engine

- Replicator yields per-barrel pose/size/count, 3D boxes and instance/semantic masks for free.
- RTX LiDAR matched to the real sensor; perfectly co-registered camera (known extrinsics).
- Scripted scenes sweep barrel count, spacing, burial depth and mutual occlusion.
- True occlusion fraction (visible/total surface points) gives a grounded x-axis for the occlusion sweep.
- Supplies the labelled training set the learned detectors require.

Camera-LiDAR fusion

- Camera-first: image detection → frustum → constrained 3D fit (e.g. Frustum-PointNet).
- LiDAR-first: 3D proposals → image verification / point painting.
- Hypothesis to test, not assume: fusion's recall advantage under occlusion vs. LiDAR-only.
- Calibration-noise injection: perturb the (sim-perfect) camera-LiDAR extrinsics to find where any fusion advantage degrades — a key real-world failure mode.

Requirement specifications & dataset request

Inputs needed to parameterise the simulation and validate on real operating conditions

Target specification

- What is the barrel size? (nominal dimensions / drum standard, e.g. 200 L)
- Are the barrels damaged or discoloured? (dents, rust, crushing; surface colour & finish)
- How much of each barrel is occluded? (burial, stacking, clutter — i.e. typical visible fraction)

Scene specification

- How many barrels per scene, and their arrangement?
- Operating environment: indoor/outdoor, substrate, illumination, airborne dust.

Sensor & data specification

- **Test-site recordings: LiDAR + camera — time-synced only if available; extrinsic calibration where available.**
- Sensor suite: LiDAR model (channels / FoV / range), camera resolution; platform & mounting (robot / drone / fixed), motion.
- Any annotations: barrel positions, counts, sizes (even approximate).

Operational requirements

- Required positional & radial accuracy, effective range, latency (online vs. offline), acceptable FP/FN trade-off.

Summary & next step

A method-agnostic benchmark with four LiDAR-only geometric detectors is operational and quantified on synthetic and real data (radius ~ 1 cm, axis $\sim 2^\circ$).

- Next: occlusion/burial sweep via Isaac Sim, learned detectors, and camera-LiDAR fusion — all on the same metrics.
- A calibrated test-site dataset would parameterise the simulation and validate under real operating conditions.